Math Inquirers

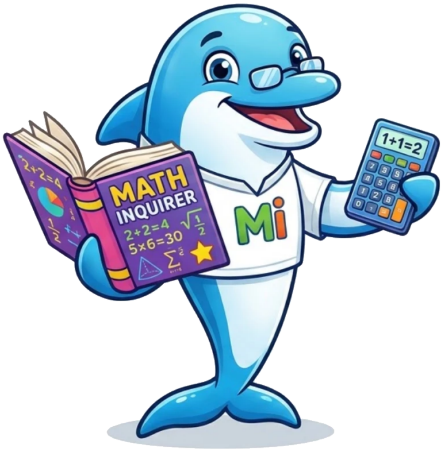
Think. Question. Prove.

Level 1 (ages 6-8) - Term 1

Tutor Answer Key

Book 2

Lessons 6 to 10



FOR THE TUTOR AND GROWN-UPS ONLY - keep away from the child

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Answer key: Lesson 6, Tens Jump, Ones Hop

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. These are Mei's sticks. She gets another bundle of 10 sticks. Fill in the blanks.

Answer: 3 bundles, 7 singles, 37 sticks

The picture shows 2 bundles and 7 singles, which is 27. One more bundle makes 3 bundles, and the 7 singles stay. 30 and 7 is 37.

Watch for: Adding the bundle to the singles (28).; Forgetting the old bundles (17).

S2. In the box, draw 3 tens as 3 tall bars and 4 single sticks. Next, draw 2 more tall bars. How many sticks are there altogether?

Answer: 54 (5 bars drawn and 4 singles)

3 bars and 4 singles is 34. Two more bars add 20, so there are 5 bars and 4 singles: 54.

Watch for: Adding the 2 to the singles (36).; Drawing 2 sticks instead of 2 bars of ten.

S3. Jump forward. Fill in the blanks.

Answer: 33, 63, 81, 67

(a) 2 tens and 1 ten make 3 tens: 33. (b) 2 tens and 4 tens make 6 tens: 63. (c) 5 tens and 3 tens make 8 tens: 81. (d) 7 is 7 ones; add 6 tens to get 67.

Watch for: Changing the ones digit (23 + 40 = 27).; (d) writing 13 (7 + 6).

S4. Jump or hop. Fill in the blanks.

Answer: 24, 38, 55, 69

(a) 3 tens take away 1 ten is 2 tens: 24. (b) 5 tens take away 2 tens is 3 tens: 38. (c) 3 is a hop: 8 ones take away 3 is 5 ones: 55. (d) 7 is a hop forward: 2 and 7 make 9 ones: 69.

Watch for: Taking the 3 from the tens digit (58 - 3 = 28).; Hopping the wrong way in (d).

S5. Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?

Answer: 38 pencils

3 boxes of ten is 30 pencils. 68 - 30 = 38: jump back 3 tens (58, 48, 38). The 8 ones stay.

Watch for: Taking away 3 (65).; Adding instead of taking away (98).

S6. Ben says 45 + 20 = 47. Fill in a to c.

Answer: No. The right answer is 65. He added 2 ones instead of 2 tens.

20 is 2 tens, so Ben should jump two tens: 45, 55, 65. He added 2 ones and got 47. A jump changes the tens digit.

Watch for: Saying he is correct because 20 has a 2 in it.; Not saying what he mixed up (any clear sentence about tens and ones is fine).

S7. Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?

Answer: 56 (two jumps of ten from 36: 46, then 56)

36 + 10 = 46, then 46 + 10 = 56. Two jumps of ten land on 56. The 6 ones never change.

Watch for: Drawing 2 small hops (38).; Changing the ones digit.

Think harder (page 3)

T1. Add 23 + 45. Fill in the blanks.

Answer: 68 (jump 40 to 63, then hop 5 to 68)

45 is 4 tens and 5 ones. 23 + 40 = 63. Then 63 + 5 = 68. Check: 2 tens and 4 tens are 6 tens; 3 ones and 5 ones are 8 ones.

Watch for: Adding only the tens and stopping at 63.; Adding 4 + 5 as 9 tens (93).

T2. Subtract 79 - 35. Fill in the blanks.

Answer: 44 (jump back 30 to 49, then hop back 5 to 44)

35 is 3 tens and 5 ones. 79 - 30 = 49. Then 49 - 5 = 44.

Watch for: Taking 35 away from the ones only.; Hopping forward instead of back.

T3. Both frogs start on 12. Frog A jumps 3 tens, then hops 4. Frog B hops 4, then jumps 3 tens. Fill in a to c.

Answer: Yes, both land on 46 (30 and 4 are added in either order)

Frog A: 12 + 30 = 42, then 42 + 4 = 46. Frog B: 12 + 4 = 16, then 16 + 30 = 46. They land on the same number: adding the same jumps and hops in a different order gives the same total.

Watch for: Saying Frog B lands on 16 because the hops came first.; Adding 3 + 4 and landing on 19.

T4. Ali starts on a number. He jumps forward 3 tens and ends on 78. Where did Ali start?

Answer: 48 (jump back 3 tens from 78)

Undo the 3 jumps of ten: 78, 68, 58, 48. Check: 48 + 30 = 78.

Watch for: Adding 30 to 78 (108).; Taking away only 3 (75).

Go further (page 4)

G1. Mei is on 7 and must get to 97. Each time she can jump 10 or jump 20. Many answers are right. Fill in a and b.

Answer: Any two different routes of 10s and 20s from 7 to 97, e.g. 7, 17, 27, 37, 47, 57, 67, 77, 87, 97 (nine jumps of 10) and 7, 27, 47, 67, 87, 97 (four of 20 and one of 10).

Every jump adds 10 or 20, and all the jumps together add 90 (97 - 7). Examples: nine jumps of 10; or 20, 20, 20, 20, 10 landing on 27, 47, 67, 87, 97. The ones digit stays 7 on every landing.

Watch for: Landing on a number that is not 10 or 20 more than the one before.; Two routes that are the same jumps in the same order.; Stopping at 87 or going past 97.

G2. Now Mei wants the fewest jumps from 7 to 97. She can jump 10 or jump 20. Fill in a and b.

Answer: 5 jumps: four jumps of 20 and one jump of 10 (any order). Four jumps travel at most 80, so she reaches only 87.

She must travel 9 tens. Four jumps of 20 make 8 tens. One jump of 10 makes the last ten. That is 5 jumps. Four jumps can travel at most 80, which only reaches 87, so 5 is the fewest.

Watch for: Saying 4 (80 only reaches 87).; Saying 9 (only using jumps of 10).

G3. Start at 25. Jump forward 3 tens, jump back 1 ten, then jump forward 2 tens. Then hop and land on 69. Fill in a to c.

Answer: 55, 45, 65, then a hop of 4 to 69. The three jumps are the same as 4 tens forward.

25 + 30 = 55. 55 - 10 = 45. 45 + 20 = 65. To land on 69 the hop is 4. Short way: 3 - 1 + 2 = 4 tens forward, so 25 + 40 = 65.

Watch for: Jumping forward on the jump back.; Saying the hop is 44 (69 - 25).; Saying 6 tens for (c) (adding 3 + 1 + 2).

Investigation (page 5): Jump road

From 6 you land on 16, 26, 36, 46, 56 and so on: every number ends in 6, so you never land on 50. Jumps of ten never change the ones digit. Only a start number that ends in 0 (0, 10, 20, 30, 40 or 50 itself) can land on 50.

Look for: A child who tests more than one start number, says the ones digit stays the same, and gives a reason using tens and ones (a jump is whole bars). A child who says start 20 can land on 50 because 20 + 30 = 50.

Exit ticket (page 5)

E1. Add.

Answer: 76

4 tens and 3 tens make 7 tens. The 6 ones stay: 76.

Watch for: Writing 49 (adding 3 to the ones).; Writing 16.

E2. Subtract.

Answer: 44 and 71

(a) 30 is 3 tens: jump back to 44. (b) 3 is a hop: hop back to 71.

Watch for: Doing the same for both parts.; (b) writing 44.

E3. Mei is on 52. She jumps forward 2 tens, then hops forward 3. Fill in the blanks.

Answer: 72 and 75

2 tens forward from 52 is 72. Then a hop of 3 forward is 75.

Watch for: Writing 54 (adding 2 to the ones).; Stopping at 72.

Sneak-peek puzzles (not in the workbook)

1. 36
2. No. 43 + 20 = 63. 20 is 2 tens, so the tens digit changes, not the ones.

Answer key: Lesson 7, Take Away, Count Up

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Here is 68 as bars and squares. Cross out 2 bars. Then fill in the blanks.

Answer: 4 tens and 8 ones are left; 68 - 20 = 48

68 is 6 tens and 8 ones. Take away 2 tens: 4 tens and 8 ones are left. That is 48. The ones did not change.

Watch for: Crossing out 2 squares instead of 2 bars (66).; Writing 84 or 4 + 8 = 12.

S2. Subtract. Use your kit if you like.

Answer: 37, 53, 56, 81

(a) 5 tens take away 2 tens is 3 tens: 37. (b) 7 ones take away 4 ones is 3 ones: 53. (c) 8 tens take away 3 tens is 5 tens: 56. (d) 6 ones take away 5 is 1: 81.

Watch for: Changing the wrong digit (57 - 20 = 55).; Taking away from both digits.

S3. Find 68 - 25. Fill in the blanks.

Answer: 48, then 43

68 take away 20 is 48. Then 48 take away 5 is 43. So 68 - 25 = 43.

Watch for: Taking away 25 as 2 + 5 = 7 (61).; Forgetting to start part (b) from 48.

S4. Subtract. Then add to check.

Answer: 34, then 46 (the same as the start, so it checks)

46 take away 10 is 36, and 36 take away 2 is 34. Check: 34 + 12 = 46, the number we started with. So 46 - 12 = 34 is right.

Watch for: Not adding back the number taken away (adding 46 + 12).; Writing 58 for the check.

S5. Mei has 45 stickers. She gives away 13. How many stickers are left? Draw a bar for each ten and a square for each one.

Answer: 32 (4 bars and 5 squares drawn; 1 bar and 3 squares crossed out)

45 is 4 tens and 5 ones. 13 is 1 ten and 3 ones. 45 take away 10 is 35. 35 take away 3 is 32. So Mei has 32 left.

Watch for: Crossing out 13 squares.; Writing 58 (adding instead of taking away).

S6. Kumar says 59 - 20 = 57.

Answer: No. 39. He took away 2 ones instead of 2 tens (57 is 59 - 2).

59 is 5 tens and 9 ones. Take away 2 tens: 3 tens and 9 ones, which is 39. Kumar took away 2 ones, so he got 57.

Watch for: Saying he is right.; Writing 79 (adding 20).

S7. A bus has 64 seats. 21 children are sitting. How many seats are empty?

Answer: 43 seats

64 take away 20 is 44. Then 44 take away 1 is 43. So 43 seats are empty.

Watch for: Adding 64 + 21 = 85.; Taking away only 2 or only 1.

Think harder (page 3)

T1. Mei has 62 beads. Ravi has 38 beads. Find how many more beads Mei has than Ravi.

Answer: 24 more beads (38 to 40 is 2, 40 to 60 is 20, 60 to 62 is 2)

38 to 40 is 2. 40 to 60 is 20. 60 to 62 is 2. The whole gap is 2 + 20 + 2 = 24. Check: 38 + 24 = 62.

Watch for: Forgetting that two tens is 20 (writing 2).; Writing 62 - 38 as 36 or 34.; Subtracting the smaller digit from the larger in each column (62 - 38 = 36).

T2. Find 83 - 77. Show how you found it.

Answer: 6 (77 to 80 is 3, 80 to 83 is 3)

77 to 80 is 3. 80 to 83 is 3. The gap is 3 + 3 = 6. Check: 77 + 6 = 83.

Watch for: Writing only one hop (3).; Taking away 83 - 7 = 76.; Subtracting the smaller digit from the larger in each column (83 - 77 = 14).

T3. Ali has 24 stickers and Ben has 31 stickers. Then each child gets 10 more stickers. Find how many more stickers Ben has than Ali now.

Answer: 7 more (the gap stays the same: 34 and 41, and 41 - 34 = 7)

Ali now has 34 and Ben has 41. The gap from 34 to 41 is 7. It was 7 before too (24 to 31), because both children got the same 10 more.

Watch for: Writing 17 (7 + 10).; Writing 41 (Ben's number).

T4. A book has 71 pages. Ravi has read 46 pages. How many pages are left?

Answer: 25 pages (46 to 50 is 4, 50 to 70 is 20, 70 to 71 is 1)

46 to 50 is 4. 50 to 70 is 20. 70 to 71 is 1. The gap is 4 + 20 + 1 = 25. Check: 46 + 25 = 71.

Watch for: Adding 71 + 46.; Forgetting the last hop of 1 (24).; Subtracting the smaller digit from the larger in each column (71 - 46 = 35).

Go further (page 4)

G1. Fill in the blanks. Circle the quicker way for each.

Answer: 71 - 68 = 3 (count up: the numbers are close). 71 - 20 = 51 (take away: 2 tens).

71 - 68: count up. 68 to 70 is 2 and 70 to 71 is 1, so 3. The numbers are close, so the gap is short. 71 - 20: take away 2 tens to get 5 tens and 1 one, which is 51. Counting up 20 would be a long way.

Watch for: Taking away 68 from 71 one at a time.; Counting up from 20 to 71 (too long).

G2. A crate has 100 eggs. Ali sells 64 of them. How many eggs are left? Find the answer two ways.

Answer: 36 eggs. Take away: 100 - 60 = 40, then 40 - 4 = 36. Count up: 64 to 70 is 6, then 70 to 100 is 30, so 36. They match.

Take away: 100 - 60 = 40, and 40 - 4 = 36. Count up: 64 to 70 is 6, and 70 to 100 is 30, so 6 + 30 = 36. Both ways give 36. Check: 64 + 36 = 100.

Watch for: Writing 164 (adding).; Writing 40 (forgetting to take away the 4 ones).

G3. Mei says 83 - 47 and 86 - 50 have the same answer.

Answer: Yes. Both are 36. Both numbers go up by 3, so the gap does not change.

83 - 47: 47 to 50 is 3, 50 to 80 is 30, 80 to 83 is 3, so 36. 86 - 50 = 36. 83 became 86 and 47 became 50: both went up by 3, so the gap between them stayed 36.

Watch for: Finding only one answer.; Saying the answers differ because the numbers differ.

Investigation (page 5): When is counting up quicker?

Counting up is quicker when the two numbers are close together, so the gap is short (52 - 49 is 3). Taking away is quicker when the number taken away is small or a whole number of tens (52 - 10 is 42). If the numbers are far apart and the small number is awkward, both ways take longer.
Look for: A child who tries several pairs, says what they notice about how far apart the numbers are, and gives a reason. Accept any clear rule that matches examples. A child who also checks by adding is doing well.

Exit ticket (page 5)

E1. Subtract.
Answer: 34
58 take away 20 is 38, and 38 take away 4 is 34.
Watch for: Writing 82 (adding).; Writing 36 (took away 2 and 2).

E2. Fill in the blanks.
Answer: 5, then 5
57 to 60 is 3, and 60 to 62 is 2. That is 5. So 62 - 57 = 5.
Watch for: Writing only the first hop (3).; Writing 15 (62 - 47).

E3. Find 90 - 87. Circle the way you used: take away or count up.
Answer: 3. Either way is fine; counting up from 87 to 90 is short because the numbers are close together.
87 to 90 is 3, so 90 - 87 = 3. Counting up is quick because the numbers are close.
Watch for: Taking away 87 from 90 one at a time.; Writing 13.

Sneak-peek puzzles (not in the workbook)

1. 11
2. 4 more. From 58 to 60 is 2, and from 60 to 62 is 2 more. 2 and 2 make 4.

Answer key: Lesson 8, Balance and Swap

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Fill in the missing numbers.

Answer: 4, 5, 3, 2

(a) 5 + 3 is 8 and 4 + 4 is 8. (b) 6 + 4 is 10 and 5 + 5 is 10. (c) 7 + 3 is 10 and 3 + 7 is 10. (d) 2 + 8 is 10 and 8 + 2 is 10.

Watch for: Writing the total of the left side in the blank (8 in part a).; Thinking the equals sign means the answer comes next.

S2. Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.

Answer: 1 more button (6 + 2 = 8 and 7 + 1 = 8)

The left plate has 6 + 2 = 8 buttons. The right plate has 7 buttons, so it needs 1 more to make 8.

Watch for: Drawing 8 more buttons on the right plate.; Forgetting the 7 buttons already there.

S3. Siti writes 6 + 3 = 5 + 9.

Answer: No. The blank should be 4 (6 + 3 = 9 and 5 + 4 = 9). Siti wrote the total of the left side, as if = meant the answer comes next.

6 + 3 is 9. The right side must also be 9. 5 and 4 more make 9, so the blank is 4. Siti wrote 9, the total of the left side, but the blank is only one of the two numbers on the right.

Watch for: Saying yes because 9 is the answer to 6 + 3.; Writing 9 in the box again (copying Siti).

S4. 7 + 5 = 12. Fill in the missing numbers.

Answer: 6, 7, 8

(a) 6 + 6 = 12. (b) 5 + 7 = 12. (c) 4 + 8 = 12. The first number goes down by 1 and the second goes up by 1, so the total stays 12.

Watch for: Leaving the second number as 5.; Making the second number go down as well.

S5. Fill in the missing numbers.

Answer: 10, 10, 30, 40

(a) 20 went up by 20 to 40, so 30 goes down by 20 to 10. (b) 50 went up by 10 to 60, so 20 goes down by 10 to 10. (c) 40 went down by 10 to 30, so 20 goes up by 10 to 30. (d) 30 went down by 10 to 20, so 30 goes up by 10 to 40.

Watch for: Writing the total of the left side (50, 70, 60 or 60).; Moving both numbers the same way (part c: 20 - 10 = 10).

S6. Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?

Answer: 5 blue pens (9 + 6 = 10 + 5)

Mei has 9 + 6 = 15 pens. Ben has 10 red pens, so he needs 5 more to make 15. Another way: his red pens are 1 more than Mei's, so his blue pens are 1 fewer: 5.

Watch for: Answering 6 (copying Mei's blue pens).; Answering 15 (the total).

S7. Fill in the missing number.

Answer: 45 (35 + 12 = 47 and 45 + 2 = 47)

The 12 went down by 10 to become 2, so the 35 goes up by 10 to become 45. Check: 35 + 12 = 47 and 45 + 2 = 47.

Watch for: Writing 47 (the total).; Writing 25 (taking a ten from the 35).

Think harder (page 3)

T1. Mei says 47 + 18 and 45 + 20 have the same total. Ali says 45 + 20 has the greater total. Do not add.

Answer: Mei. 47 went down by 2 to 45 and 18 went up by 2 to 20. The 2 moved across, so both totals are 65.

47 to 45 is 2 less. 18 to 20 is 2 more. The 2 moved from one number to the other, so the total stays the same. Both sums make 65.

Watch for: Saying 45 + 20 is bigger because 20 is a bigger number than 18.; Adding both sums and not saying how the numbers moved (correct, but check the reason).

T2. Do not add. Circle LEFT or RIGHT for the greater total, or circle SAME.

Answer: SAME, SAME, RIGHT

(a) 29 up 1 to 30, 14 down 1 to 13: SAME (both 43). (b) 53 down 3 to 50, 20 up 3 to 23: SAME (both 73). (c) 26 up 2 to 28 and 15 up 1 to 16: both went up, so RIGHT is bigger (41 and 44).

Watch for: Saying SAME for part c because both numbers changed.; Looking only at the biggest numbers.

T3. Fill in the missing number. Then find the total.

Answer: 25, then the total is 75

49 went up by 1 to 50, so 26 goes down by 1 to 25. 50 + 25 = 75, so 49 + 26 = 75.

Watch for: Writing 27 (making 26 go up as well).; Writing 75 in the blank.

T4. Ben writes 24 + 17 = 26 + 19.

Answer: No. 24 + 17 = 41 but 26 + 19 = 45. Both numbers went up by 2. Change 19 to 15.

24 went up by 2 to 26, so 17 should go down by 2 to 15. Ben made it go up by 2 to 19, so the right side is 4 too big. Check: 24 + 17 = 41 and 26 + 15 = 41.

Watch for: Saying it is balanced because both numbers changed by 2.; Changing the 26 and not saying which number you change.

Go further (page 4)

G1. Fill in the missing numbers so that all three sums have the same total.

Answer: 27 and 17 (every sum is 47)

19 up 1 to 20, so 28 down 1 to 27. The total is 20 + 27 = 47 (check: 19 + 28 = 47). Then [] + 30 = 47, so the second blank is 47 - 30 = 17.

Watch for: Writing 27 in both blanks.; Writing 30 or 28 as the second blank.

G2. Fill in the blanks so both sides are the same. Write 3 different pairs. Many answers are right.

Answer: Any pair where the first blank is 4 more than the second, for example 5 and 1, 6 and 2, 7 and 3 (also 4 and 0).

If the second blank is 1, the right side is 10, so the first blank is 5. If it is 2, the right side is 11, so the first blank is 6. The first blank is always 4 more than the second, because 9 is 4 more than 5.

Watch for: Writing the same number in both blanks.; Giving one pair only.

G3. Mei has 28 stickers on a page and 17 in her pocket. Ali, Ben and Kumar have the same total as Mei. Fill in the blanks.

Answer: 15, 20 and 33 (Mei has 45 stickers; 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45)

Mei has 28 + 17 = 45. Ali: 30 is 2 more than 28, so his pocket has 2 fewer than 17: 15. Ben: 25 is 3 fewer than 28, so his pocket has 3 more than 17: 20. Kumar: his pocket 12 is 5 fewer than 17, so his page has 5 more than 28: 33. Check: 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45.

Watch for: Writing 17 for Ali's pocket.; Making Ben's pocket 14 (moving the wrong way).; Writing 17 or 45 for Kumar's page.

Investigation (page 5): Keep it the same

First step: 7 + 5 is not 10 (the totals are 10, 10 and 12). From 6 + 4 to 7 + 3 the 4 goes down 1. Next: add some to one number and take the same from the other and the total stays the same (give and take). Add 3 to both: the total goes up by 6. Take 2 from both: the total goes down by 4. When both numbers change the same way, the total changes by double.

Look for: A child who tries more than one pair of numbers, says it stays the same when one goes up and the other goes down by the same amount, and notices that changing both numbers the same way changes the total by double. Any clear reason is fine.

Exit ticket (page 5)

E1. Fill in the missing number.

Answer: 4

9 went up by 1 to 10, so 5 goes down by 1 to 4. Both sides make 14.

Watch for: Writing 14 (the total).; Writing 6 (making the 5 go up).

E2. Is 28 + 6 = 30 + 4 balanced?

Answer: Yes

28 went up by 2 to 30 and 6 went down by 2 to 4. Both sides make 34.

Watch for: Writing no because the numbers look different.

E3. Fill in the missing number.

Answer: 2. The 30 went up by 10 to 40, so the 12 goes down by 10 to 2.

30 went up by 10, so 12 goes down by 10 to 2. Both sides make 42.

Watch for: Writing 42 (the total).; Writing 22 (making the 12 go up).

Sneak-peek puzzles (not in the workbook)

1. Yes, the same: 10 beads each.
2. 4. 9 is 1 more than 8, so the blank is 1 less than 5. (Both sides make 13.)

Answer key: Lesson 9, Draw and Measure

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. In the box, draw a line 7 cm long. Start at the 0 mark.

Answer: The line ends at the 7 mark (it starts at the 0 mark and is 7 cm long)

Put a dot for the start. Line up the 0 mark with the dot. Make a small mark at the 7. Join the dot and the mark with a straight line. The line ends at the 7 mark, so it is 7 cm long.

Watch for: Starting at the edge of the ruler instead of the 0 mark (the line comes out a little too long).; Drawing to the 8 mark because the 0 was counted as one.

S2. In the box, draw a line 9 cm long. Ali's line is 4 cm long.

Answer: 5 cm longer (9 take away 4)

Your line is 9 cm and Ali's line is 4 cm. Count on from 4 to 9: that is 5 jumps. So your line is 5 cm longer (9 take away 4 is 5).

Watch for: Writing 13 by adding the two lengths.; Starting the 9 cm line at a mark other than 0.; Writing 9 (the length of the line you drew).

S3. Measure something short from your table, such as a crayon or a rubber. Write your guess first. Then measure it in whole cm.

Answer: Any answer that matches the thing you measured and is close to your guess. A grown-up checks.

A grown-up checks: put one end on the 0 mark and read the mark at the other end. A crayon is about 8 to 9 cm and a rubber about 5 to 6 cm. Accept any whole number within 1 cm of the tip.

Watch for: Starting at the edge of the ruler instead of the 0 mark.; Writing the guess in the measure space, or leaving off the unit.

S4. Look at the ruler. Ali's pencil and Mei's pencil both start at the 0 mark.

Answer: 13 cm; 9 cm; Ali's is 4 cm longer

Both start at the 0 mark, so each tip mark is the length: Ali 13 cm, Mei 9 cm. From 9 to 13 is 4 jumps (13 take away 9 is 4), so Ali's pencil is 4 cm longer.

Watch for: Adding 13 + 9 = 22.; Writing 9 as the difference (that is the length of Mei's pencil).

S5. Mei's line is 5 cm long. In the box, draw a line 3 cm longer than Mei's line.

Answer: The second line is 8 cm long (5 + 3)

Mei's line is 5 cm. 3 cm longer is 5 + 3 = 8 cm. Draw it from the 0 mark to the 8 mark.

Watch for: Drawing a line 3 cm long.; Starting the line at the edge of the ruler, so it is a little too long.

S6. Look at Mei's line on the ruler. Mei says it is 8 cm long.

Answer: No. Her line is 7 cm. She should draw on to the 8 mark

Mei started at 0 and stopped at the 7 mark, so her line is 7 cm long. A line 8 cm long must end at the 8 mark. She should draw on to the 8 mark.

Watch for: Saying yes because Mei said 8.; Counting the 0 as one, so the 7 mark looked like 8.

Think harder (page 3)

T1. Use your ruler to draw a path with 3 straight pieces: 5 cm, 3 cm and 4 cm.

Answer: 12 cm (5 + 3 + 4)

Draw 5 cm, turn, draw 3 cm, turn, draw 4 cm. The path is as long as all three pieces together: 5 + 3 = 8, and 8 + 4 = 12 cm.

Watch for: Measuring in a straight line from the start to the end of the path.; Drawing one long line instead of turning at the corners.

T2. Look at line A on the ruler. Line B starts at the 3 mark and is 6 cm long.

Answer: 7 cm; the 9 mark

(a) From the 2 mark to the 9 mark is 7 jumps: 9 take away 2 is 7 cm, not 9 cm. (b) Start at the 3 mark and jump on 6: 3 + 6 = 9. The line ends at the 9 mark.

Watch for: Writing 9 for line A because the end is at the 9 mark.; Taking 6 away from 3, or writing 6 for line B.

T3. Look at the ruler. Both pencils start at the 0 mark. Each tip is between two marks.

Answer: 9 cm; 13 cm; 4 cm longer

Ali's tip is just past the 9 mark, so it is closer to 9: 9 cm. Mei's tip is just before the 13 mark, so it is closer to 13: 13 cm. From 9 to 13 is 4 jumps, so Mei's pencil is 4 cm longer.

Watch for: Choosing the mark the tip has not reached (12 for Mei).; Writing 10 for Ali because the tip looks nearly there.; Writing 3 by using 12 and 9.

Go further (page 4)

G1. A ribbon is 30 cm long. Mei cuts all of it into 3 pieces: A, B and C. Piece A is 8 cm long. Piece B is 3 cm longer than piece A. How long is piece C?

Answer: Piece C is 11 cm (B is 11 cm, A and B together are 19 cm, and 30 take away 19 is 11)

B is 8 + 3 = 11 cm. A and B together are 8 + 11 = 19 cm. The ribbon is 30 cm, so C is 30 take away 19, which is 11 cm. Check: 8 + 11 + 11 = 30.

Watch for: Taking 30 - 8 = 22 and stopping.; Writing 3 for piece B instead of 3 more than A.

G2. A snail crawls along a long ruler from the 4 mark to the 13 mark. Then it crawls back 5 cm.

Answer: At the 8 mark; it crawled 14 cm in all (9 cm forward and 5 cm back)

Forward from the 4 mark to the 13 mark is 9 cm. Back 5 cm: 13 take away 5 puts the snail at the 8 mark. In all it crawled 9 + 5 = 14 cm, although it is only 4 cm from where it began.

Watch for: Writing 9 for the total and forgetting the way back.; Writing 4 as the mark (the distance from the start, not the mark).

G3. Draw a path with 3 straight pieces. The pieces are all different lengths, in whole cm. The whole path is exactly 20 cm long. Many answers are right.

Answer: Many answers are right: any 3 different whole numbers that add up to 20, for example 5, 7 and 8 (5 + 7 + 8 = 20).

One answer is 5, 7 and 8: 5 + 7 is 12 and 12 + 8 is 20. Other answers are 3, 8 and 9, or 4, 7 and 9. If the total is not 20, change one piece until it is.

Watch for: Two pieces the same length (for example 5, 5 and 10).; A total of 19 or 21 because a piece was drawn a little too long or too short.

Investigation (page 5): Does the start mark matter?

The length stays 6 cm every time. From the 2 mark the line ends at 8; from 3 it ends at 9; from 5 it ends at 11. The end mark is always the start mark plus 6, so the end mark take away the start mark is always 6. Reading only the end mark gives the length only when the line starts at 0.

Look for: A child who tries more than one start mark, writes both marks, and says that the end mark take away the start mark (or the number of jumps) is the length. A 15 cm ruler allows start marks up to 9.

Exit ticket (page 5)

E1. In the box, draw a line 6 cm long. Start at the 0 mark.

Answer: A line from the 0 mark to the 6 mark

Put the 0 mark of the ruler on the dot and draw to the 6 mark. The line ends at the 6 mark.

Watch for: Starting at the edge of the ruler.; Ending at the 7 mark after counting the 0 as one.

E2. Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.

Answer: 9 cm longer

The rubber is 6 cm and the pencil is 15 cm. 15 take away 6 is 9, so the pencil is 9 cm longer.

Watch for: Adding: 21.; Writing 15 (the pencil, not the difference).

E3. A line starts at the 3 mark and ends at the 10 mark. How long is the line?

Answer: 7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10)

Count the jumps from 3 to 10: 7 jumps. Or take 10 take away 3, which is 7. The line is 7 cm long, not 10 cm.

Watch for: Writing 10 (the end mark).; Writing 13 by adding.

Sneak-peek puzzles (not in the workbook)

1. A line from the dot that is 3 cm long: the 0 mark on the dot and the 3 mark at the other end.
2. No. It is 8 cm long. The pencil starts at the 1 mark, not the 0 mark, so the first centimetre of the ruler is not part of the pencil (9 take away 1 is 8).

Answer key: Lesson 10, Story Steps: Read It Twice

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Ben has 30 cards. He gives 20 cards away. After that, he gets 6 more cards.

Answer: Step 1: 30 - 20 = 10. Step 2: 10 + 6 = 16. Ben has 16 cards.

First, Ben gives away 20, so we take away: 30 - 20 = 10. Then he gets 6 more, so we add: 10 + 6 = 16. Ben has 16 cards now.

Watch for: Adding the 20 (30 + 20) because 20 is in the story.; Using 30 again in Step 2 instead of the 10 left after Step 1.

S2. Mei has 42 stickers. She gets 7 more stickers. Then she gives 10 stickers to Ali. Write one number sentence for each step.

Answer: Step 1: 42 + 7 = 49. Step 2: 49 - 10 = 39. Mei has 39 stickers.

Step 1: she gets 7 more, so add: 42 + 7 = 49. Step 2: she gives 10 away, so take away: 49 - 10 = 39. Mei has 39 stickers now.

Watch for: Using 42 again in Step 2 (42 - 10 = 32).; Adding both numbers (42 + 7 + 10).

S3. Ravi has 25 pencils. He buys 10 more pencils. Then he loses 3 pencils. Circle the 2 number sentences that fit the story.

Answer: Circle 25 + 10 = 35 and 35 - 3 = 32. Ravi has 32 pencils.

Step 1: buys 10 more, so 25 + 10 = 35. Step 2: loses 3, so 35 - 3 = 32. The sentences 25 - 10 = 15 and 35 + 3 = 38 use the wrong sign.

Watch for: Circling 25 - 3 = 22 because both numbers are in the story.; Circling a sentence with the wrong sign.

S4. Write + or - in each box.

Answer: +, -, -, +

Finds more means add (+). Eats means take away (-). Gives away means take away (-). Join means more, so add (+).

Watch for: Choosing + for gives away because there is a number in the sentence.; Choosing - for join.

S5. Ali has 23 marbles. Ben has 10 more than Ali. How many marbles do they have altogether? Use counters if you like.

Answer: Step 1: Ben has 23 + 10 = 33. Step 2: 23 + 33 = 56. They have 56 marbles.

Step 1: Ben has 10 more than Ali, so 23 + 10 = 33. Step 2: altogether means add Ali and Ben: 23 + 33 = 56 marbles.

Watch for: Answering 33 (stopping after Step 1).; Adding 23 + 10 and calling that altogether.

S6. A bus has 30 children. 10 children get off. Then 4 children get on. Siti writes 30 + 10 = 40 for Step 1.

Answer: No. Get off means take away: 30 - 10 = 20. Then 20 + 4 = 24. There are 24 children.

Children who get off leave the bus, so Step 1 is a take away: 30 - 10 = 20. Siti added the 10. Then 4 get on, so add: 20 + 4 = 24.

Watch for: Circling Yes.; Adding both numbers to 30 (30 + 10 + 4 = 44).

Think harder (page 3)

T1. Ravi has 15 marbles. He has 3 more marbles than Mei.

Answer: Mei has 12 marbles. Together they have 27.

Ravi has 3 more than Mei, so Mei has 3 fewer than Ravi: 15 - 3 = 12. Together: 15 + 12 = 27 marbles.

Watch for: Adding 15 + 3 = 18 because the word more is there.; Answering 12 for the second part.

T2. Mei is in a line to buy lunch. There are 6 children in front of her and 4 children behind her. How many children are in the line?

Answer: 11 children (6 + 4 + 1, counting Mei).

The children in front are 6 and behind are 4: 6 + 4 = 10. Mei is in the line as well, so 10 + 1 = 11 children.

Watch for: Answering 10 and leaving Mei out.

T3. Ben has 12 marbles. Mei has 5 more marbles than Ben. Siti has 3 fewer marbles than Mei. How many marbles does Siti have?

Answer: Mei has 17. Siti has 14 marbles.

Step 1: Mei has 5 more than Ben: 12 + 5 = 17. Step 2: Siti has 3 fewer than Mei: 17 - 3 = 14 marbles.

Watch for: Taking 3 from 12 (Siti is compared with Mei, not Ben).; Answering 17 (stopping after Step 1).

Go further (page 4)

G1. A shop has 60 pencils. It sells 24 pencils on Monday and 15 pencils on Tuesday. On Wednesday, a box of 30 pencils comes in. Write one number sentence for each step.

Answer: Step 1: 60 - 24 = 36. Step 2: 36 - 15 = 21. Step 3: 21 + 30 = 51. The shop has 51 pencils.

Monday: 60 - 24 = 36. Tuesday: 36 - 15 = 21. Wednesday a box comes in, so add: 21 + 30 = 51 pencils.

Watch for: Adding 24 and 15 and taking that from 60 but forgetting the box.; Starting each step from 60 again.

G2. Step 1: 25 + 10 = 35. Step 2: 35 - 8 = 27. Write a story that fits both steps. End your story with a question. Many stories are right.

Answer: Any story where 25 grows by 10 and then 8 are taken away, and the question asks how many now. The answer is 27.

For example: Ali has 25 stickers. He gets 10 more. Then he gives 8 to his sister. How many stickers does Ali have now? 25 + 10 = 35, 35 - 8 = 27.

Watch for: A story where the second step starts from 25 instead of 35.; No question at the end.

G3. Mei, Ali and Ben have 70 shells altogether. Mei has 25 shells. Ali has 10 more shells than Mei. Write one number sentence for each step. How many shells does Ben have?

Answer: Ali has 35. Mei and Ali have 60. Ben has 10 shells.

Step 1: Ali has 25 + 10 = 35. Step 2: Mei and Ali have 25 + 35 = 60. Step 3: Ben has the rest: 70 - 60 = 10 shells.

Watch for: Answering 35 or 60 (stopping early).; Taking only Mei's shells from 70.

Investigation (page 5): Does the order matter?

Both orders give the same answer (20 + 10 - 5 = 25 and 20 - 5 + 10 = 25) whenever both orders can be done. But the order can matter for what you are allowed to do: from 4 you cannot take away 6 first, though 4 + 10 - 6 = 8 works. So: the answer is the same, but you cannot take away more than you have.

Look for: A child who tests several numbers, says what they notice, and finds a case where taking away first is not possible (start 4, take away 6).

Exit ticket (page 5)

E1. A bus has 25 children. 10 children get off. Then 3 children get on. How many children are on the bus now?

Answer: 18 (25 - 10 = 15, then 15 + 3 = 18)

Get off means take away: 25 - 10 = 15. Get on means add: 15 + 3 = 18.

Watch for: Writing 38 (adding the 10).; Writing 15 (stopping after Step 1).

E2. Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers does Mei have?

Answer: 11 (6 + 5)

Mei has more than Ben, so add: 6 + 5 = 11.

Watch for: Writing 1 (taking away).; Writing 5.

E3. Siti has 20 beads. Ali has 7 fewer beads than Siti. How many beads does Ali have?

Answer: 13 (20 - 7)

Ali has fewer, so take away: 20 - 7 = 13.

Watch for: Writing 27 (adding because the number 7 is there).; Writing 7.

Sneak-peek puzzles (not in the workbook)

1. 7
2. 11. First find Ben's marbles: 4 + 3 = 7. Then add Ali's 4: 4 + 7 = 11.